

Lab 8 Activity

1

Load in the `Ornstein` data from the `carData` package, along with the `tidyverse` package. Read the help page for the `Ornstein` data to better understand these data.

```
library(tidyverse)
data("Ornstein", package = "carData")
```

For today, we will use `interlocks` as the DV, and `sector` and `nation` as IVs. However, we are only interested in whether the nation is Canada or not, so we want to combine the `OTH`, `UK`, and `US` factor levels. One way to accomplish this is to run the following code. (Then use `canada` as an IV instead of `nation`).

```
Ornstein <- Ornstein %>%
  mutate(canada = factor(nation == "CAN"))
```

Finally, we will omit data from the “BNK”, “CON”, and “HLD” levels of `sector` because they have relatively few observations.

```
Ornstein <- Ornstein %>%
  filter(!sector %in% c("BNK", "CON", "HLD"), .preserve = FALSE) %>%
  mutate(sector = factor(sector))
```

2

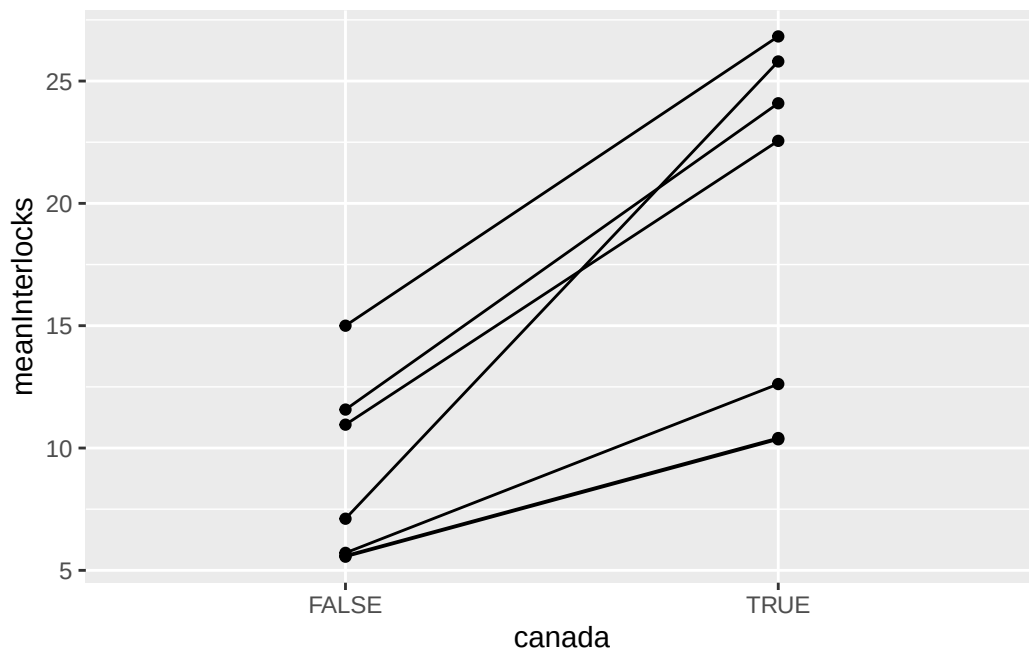
Conduct graphical and numerical exploratory analyses on these data. How do these analyses inform how you set up the ANOVA and what results you expect to find?

```
# quantitative: we want to check for balanced data among our IVs
table(Ornstein$sector, Ornstein$canada) # unbalanced data - should use type II or III SS
```

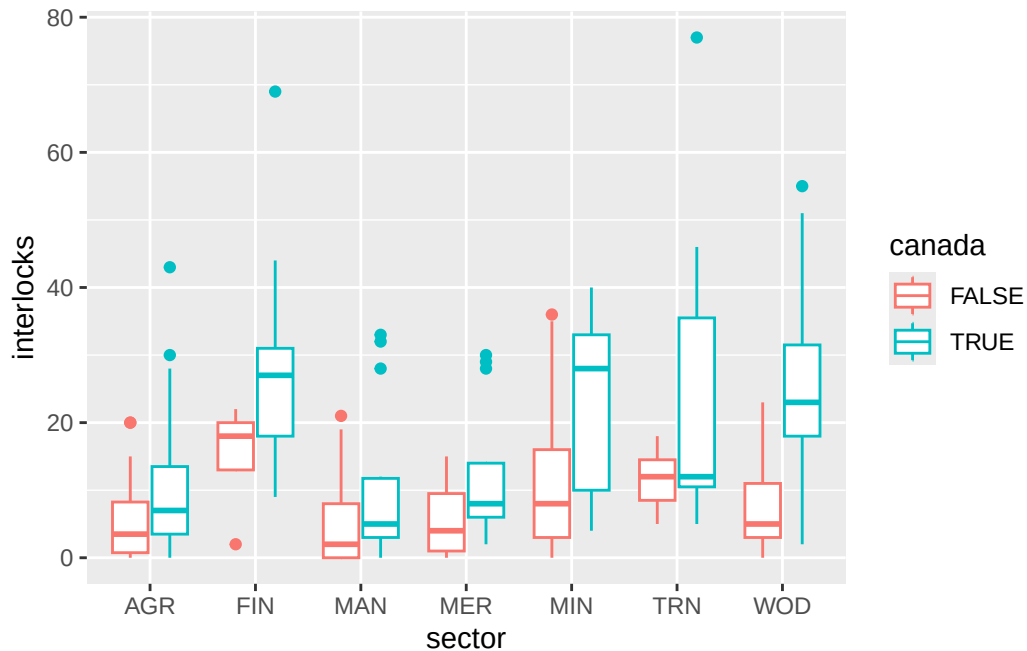
	FALSE	TRUE
AGR	20	27
FIN	5	17
MAN	34	14
MER	7	13
MIN	45	9
TRN	7	11
WOD	9	10

```
# graphical: look at interaction plot
Ornstein2 <- Ornstein %>% group_by(sector, canada) %>%
  summarize(meanInterlocks = mean(interlocks), .groups = "keep")

ggplot(Ornstein2, aes(canada, meanInterlocks, group = sector)) +
  geom_point() + geom_path() # some differences in slope, might expect an interaction - use t
```



```
# graphical: look at boxplots for homogeneity of variance
ggplot(Ornstein, aes(sector, interlocks, col = canada)) +
  geom_boxplot() # some major differences in the shape of the boxplots, probably want to not
```



3

Conduct a two-way ANOVA on these data using options informed by the previous step, and calculate effect sizes. What do you find?

```
library(ez) # use ezANOVA for correct type III SS
Ornstein <- Ornstein %>% mutate(ID = factor(1:nrow(Ornstein))) # create ID variable

# White's adjustment
mod1 <- ezANOVA(data = Ornstein, dv = interlocks, wid = ID,
  between = c(sector, canada), white.adjust = TRUE,
  type = 3, return_aov = TRUE)
```

Warning: Data is unbalanced (unequal N per group). Make sure you specified a well-considered value for the type argument to ezANOVA().

Coefficient covariances computed by hccm()

```
mod1 # no significant interaction, but both main effects are significant
```

```
$ANOVA
```

	Effect	DFn	DFd	F	p	p<.05
2	sector	6	214	5.839269	1.173870e-05	*
3	canada	1	214	28.120317	2.834722e-07	*
4	sector:canada	6	214	1.134446	3.433283e-01	

```
$`Levene's Test for Homogeneity of Variance`
```

	DFn	DFd	SSn	SSd	F	p	p<.05
1	13	214	1792.91	15190.56	1.942921	0.02699431	*

```
$aov
```

```
Call:
```

```
aov(formula = formula(aov_formula), data = data)
```

```
Terms:
```

	sector	canada	sector:canada	Residuals
Sum of Squares	6558.523	3609.973	974.841	25679.137
Deg. of Freedom	6	1	6	214

```
Residual standard error: 10.95427
```

```
Estimated effects may be unbalanced
```

```
# probably unwise to use WLS since that requires aov, which doesn't give us an easy option f
```

```
# effect sizes
```

```
library(sjstats)
```

```
anova_stats(mod1$aov)
```

```
etasq | partial.etasq | omegasq | partial.omegasq | epsilonsq | cohens.f
```

0.178		0.203		0.158		0.176		0.159		0.505
0.098		0.123		0.094		0.113		0.095		0.375
0.026		0.037		0.007		0.009		0.007		0.195

```
etasq | term | sumsq | df | meansq | statistic | p.value | power
```

0.178		sector		6558.523		6		1093.087		9.109		< .001		1.000
0.098		canada		3609.973		1		3609.973		30.084		< .001		1.000

```
0.026 | sector:canada | 974.841 | 6 | 162.473 | 1.354 | 0.235 | 0.540
      | Residuals | 25679.137 | 214 | 119.996 | | |
```

```
# relatively small effects for both factors, sector has a larger effect than canada
```

4

Based on your results in the previous part, calculate either simple main effects or ordinary main effects. What do you conclude?

```
# want to do ordinary main effects since the interaction is not significant

Ornstein %>% group_by(canada) %>% summarize(mean(interlocks))
```

```
# A tibble: 2 x 2
  canada `mean(interlocks)`
  <fct>      <dbl>
1 FALSE      8.30
2 TRUE     17.5
```

```
# Canadian companies have more interlocks on average than non-Canadian companies (makes sense)

# for sector, use any post-hoc test for a one-way ANOVA (pairwise t-test with holm shown here)
pairwise.t.test(x = Ornstein$interlocks, g = Ornstein$sector,
                p.adjust.method = "holm")
```

Pairwise comparisons using t tests with pooled SD

data: Ornstein\$interlocks and Ornstein\$sector

	AGR	FIN	MAN	MER	MIN	TRN
FIN	8.3e-06	-	-	-	-	-
MAN	1.0000	7.9e-07	-	-	-	-
MER	1.0000	0.0029	1.0000	-	-	-
MIN	0.5279	0.0034	0.1469	1.0000	-	-
TRN	0.0153	1.0000	0.0034	0.2220	0.5279	-
WOD	0.1050	0.5279	0.0279	0.5860	1.0000	1.0000

P value adjustment method: holm

```
# significant differences between FIN vs. AGR, MAN, MER, and MIN
#   also between TRN vs. AGR and MAN
```

```
Ornstein %>% group_by(sector) %>% summarize(mean(interlocks))
```

```
# A tibble: 7 x 2
  sector `mean(interlocks)`
  <fct>      <dbl>
1 AGR         8.36
2 FIN        24.1
3 MAN         6.96
4 MER        10.2
5 MIN        12.9
6 TRN        19.2
7 WOD        16.9
```

```
# FIN has more interlocks than most other, TRN also, to some extent
```